

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

April 2013

EC207 Control Systems

Date: 24.04.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

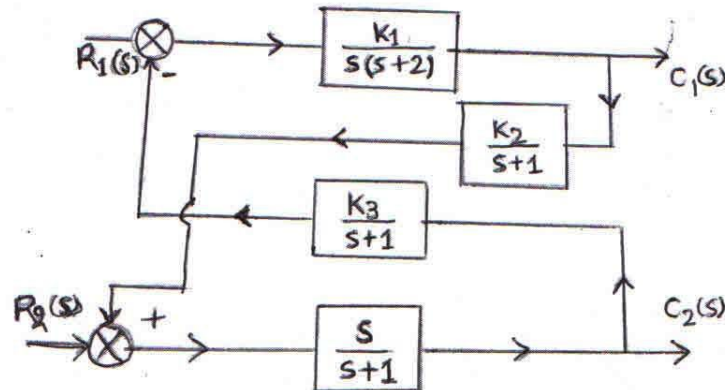
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use Graph Papers.

## SECTION – I

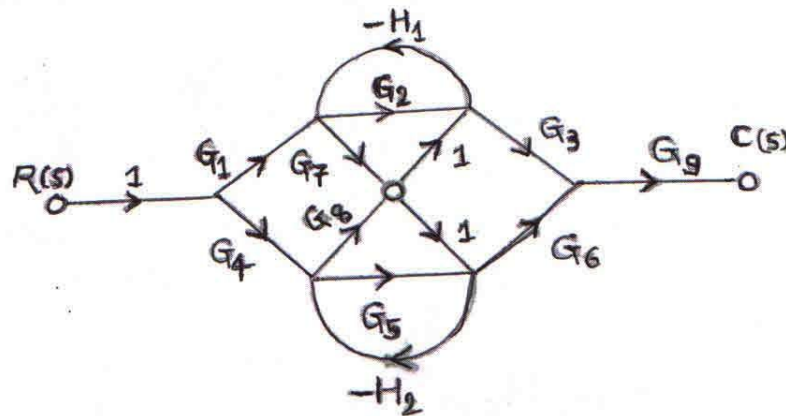
Q - 1 (a) Give the differences between block diagram representation and signal flow graph representation of system. [04]

(b) Discuss the difference between open loop control system and close loop control system. [03]

Q - 2 (a) Determine  $\frac{C_1(s)}{R_1(s)}$  and  $\frac{C_2(s)}{R_2(s)}$  for multiple input systems shown below using block diagram reduction techniques. [07]

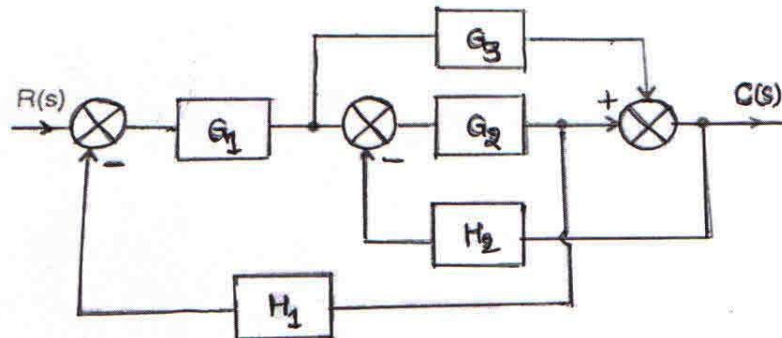


(b) Using Mason's gain formula, find  $\frac{C(s)}{R(s)}$  of the signal flow graph shown in figure. [07]



**OR**

- Q - 2 (a) Evaluate the transfer function  $\frac{C(s)}{R(s)}$  for the system shown in figure below, using block diagram reduction techniques. [08]



- (b) A system is described by [06]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

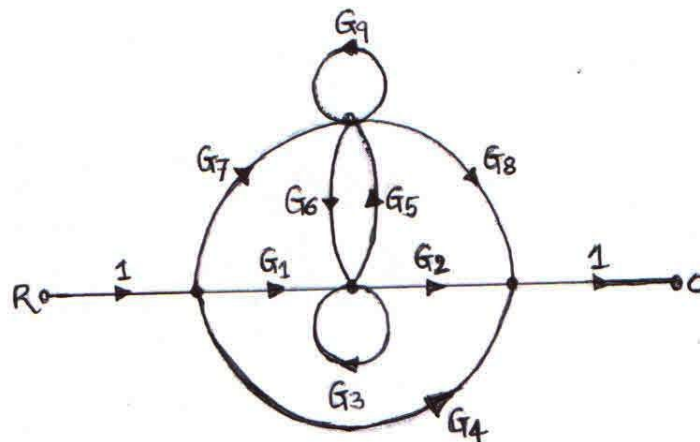
Test its state controllability and observability.

- Q - 3 (a) The open-loop transfer function of a feedback control system is [07]

$$G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$

Find the Root Locus as K is varied from 0 to  $\infty$ .

- (b) Using Mason's gain formula, find  $\frac{C}{R}$  of the signal flow graph shown in figure. [07]



**OR**

- Q - 3 (a)** Consider a unity feedback control system with an open-loop transfer function [07]

$$G(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the Root Locus of the system with the gain K as a variable.

- (b)** Write short note on Synchro transmitter and receiver. [07]

**SECTION - II**

- Q - 4 (a)** Sketch the asymptotic bode plot for the transfer function [07]

$$G(s)H(s) = \frac{10(s+5)}{s(s+0.5)(s+1)}$$

Comment on Stability.

- Q - 5 (a)** A unity feedback control system has [07]

$$G(s)H(s) = \frac{80}{s(s+2)(s+20)}$$

Draw the bode plot.

- (b)** Consider characteristics equation of a closed-loop control system as [07]

$$s^3 + 3Ks^2 + (K+2)s + 4 = 0$$

Check its stability and find the range of K so that system is stable.

**OR**

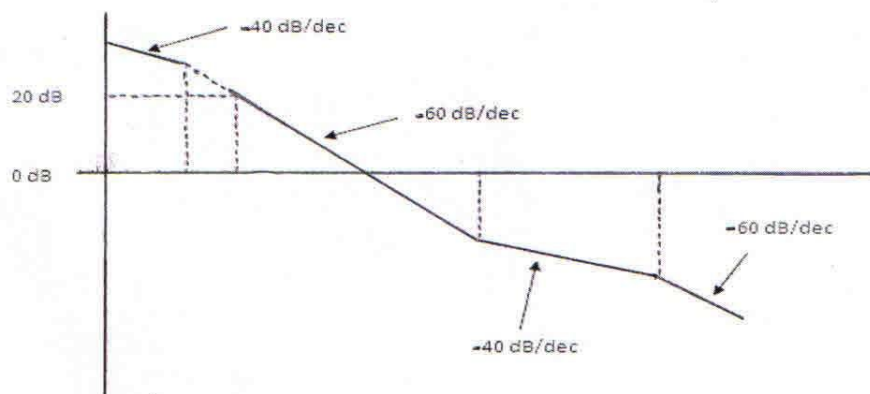
- Q - 5 (a)** Using Routh's test determine the stability of a system whose characteristics equation is given by [06]

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

- (b)** Using Nyquist criterion, examine the closed-loop stability of a system whose open-loop transfer function is [08]

$$G(s)H(s) = \frac{50}{(s+1)(s+2)}$$

- Q - 6 (a)** Determine the transfer function from the magnitude plot shown below. [06]



(b) Sketch the Polar plot for a system with

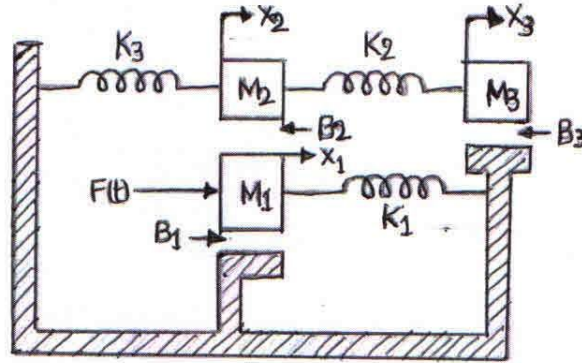
[08]

$$G(s)H(s) = \frac{10}{s(s+1)(s+2)}$$

Calculate its gain margin in dB and comment on stability.

OR

Q - 6 (a) (I) For the following translation system shown in Figure, Obtain the differential equations and draw the electrical network using force-voltage analogy. [08]



(b) A unity feedback system has open loop transfer function given below

[06]

$$G(s) = \frac{(1 + 0.4s)}{s(s + 0.6)}$$

- 1) Obtain characteristic equation.
- 2) Obtain unit step response.
- 3) What is rise time?
- 4) What is peak overshoot?
- 5) Find out settling time for the system.

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